Republic of the Philippines

Isabela State University

Echague Campus

COURSE SYLLABUS

First Semester, AY 2026-2027

Course Number	Course	Credit Units	Hours per Week
LM INST 4	Data Science Analytics	2 units	1 lec / 3 lab hours

VISION

A model SMART-Green University with a global reputation for positive societal transformation.

MISSION

The Isabela State University is committed to developing lifelong and future-ready professionals empowered to become generators of knowledge and agents of positive change for the inclusive and sustainable transformation of communities through a SMART-Green higher education experience.

QUALITY POLICY

The Isabela State University is determined to be a lead university in instruction, research, extension and resource generation through continuous improvement of services and commits to comply with the standards set by statutory, regulatory and accrediting bodies. To uphold this commitment, ISU shall attain the following quality objectives:

1. Sustain academic excellence and quality instruction;
2. Generate research work breakthroughs;
3. Engage in sectoral activities for communal development;
4. Develop products for globalization;
5. Support students' participation to local and international fora to enhance their potentialities.

INSTITUTIONAL OUTCOMES

IO1 - Innovator	Research-driven, technologically adept global citizens.
IO2 - People-Oriented	Ethical, socially responsible servant-leaders.
IO3 - Entrepreneur	Opportunity-seeking and value-creating professionals.
IO4 - Naturalist	Stewards of the environment and sustainable development.

ISABELA STATE UNIVERSITY EXPECTED GRADUATE ATTRIBUTES

Code	Graduate Attribute	Code	Graduate Attribute
GA1	Critical and creative thinker	GA6	Service-oriented leader and team member
GA2	Research-informed and technology-enabled innovator	GA7	Opportunity-seeking problem solver
GA3	Socially responsible citizen	GA8	Entrepreneurial and intrapreneurial actor
GA4	Ethical professional	GA9	Environmentally conscious individual
GA5	Socially responsible citizen	GA10	Sustainability-oriented practitioner

COURSE DESCRIPTION

This course introduces Bachelor of Science in Legal Management students to the fundamental principles of data science and analytics and to how data supports evidence-based decision-making in legal practice, compliance, and organizational management. It covers data collection, cleaning, and preparation; descriptive analytics; data exploration and visualization; and an introduction to predictive and prescriptive analytics, using Microsoft Excel and data relevant to legal and management settings, such as case loads, compliance records, contract and transaction logs, and client or organizational data. The course develops students' ability to apply legal and management principles together with ICT and decision-making tools, to

communicate data-driven insights, and to observe ethical, legal, and data privacy responsibilities in handling information, consistent with the outcomes-based education standards of CHED Memorandum Order No. 46, s. 2012 and with the legal and management orientation of the BS Legal Management program.

Prerequisite/s: None

COURSE MAPPING

Goals of the College

- Equip students with the conceptual, human, and technical knowledge, skills, and competence in law and management for their complex functions as responsible professionals and citizens in the world of work.
- Provide students with professional training through immersion and practicum to prepare them for the complex functions of legal support, compliance, and organizational administration.
- Equip students with contemporary and relevant legal and management concepts, and the tools and techniques necessary to enhance good governance and the rule of law.
- Prepare students to become efficient, effective, and ethical legal managers and administrators who are fully aware of their social responsibility, not only to their organizations but also in service to the community.
- Provide future professionals with skills in research, policy analysis, and decision making in different areas of legal and management practice.

Program Outcomes (BS Legal Management)

PO1	Apply legal and management principles, theories, and methods to analyze problems in various types of organizations.
PO2	Assess complex legal and business problems using appropriate legal research, analytical, and management tools, techniques, and frameworks.
PO3	Apply information and communication technology (ICT) skills, including data management and analysis tools, as required by legal and business environments.
PO4	Demonstrate commitment to ethical, legal, and professional responsibility in the practice of legal management.

COURSE LEARNING OUTCOMES (CLOS)

CLO1	Explain the concepts and processes of data science and data analytics and their applications in legal practice and management, using fundamental MS Excel tools.
CLO2	Collect, clean, and prepare legal and management data in MS Excel and summarize it using descriptive analytics, observing data ethics, confidentiality, and data privacy principles.
CLO3	Create data visualizations and interactive dashboards in MS Excel that communicate legal and management insights to support organizational decision-making.
CLO4	Apply predictive and prescriptive analytics (regression, time-series forecasting, and what-if analysis) in MS Excel to forecast outcomes and recommend data-driven, ethically sound actions for legal and management problems.

ALIGNMENT MATRIX 1. LILO - CLO - PO MAPPING

Weeks	Lesson Intended Learning Outcomes (LILOs)	Aligned CLO	PO1	PO2	PO3	PO4
1	State the vision, mission, goals, and institutional outcomes of ISU; relate the course learning outcomes to the program outcomes; practice university rules and regulations, including the Classroom AI Policy.	Orientation / Preparatory				
2-3	Differentiate data science and data analytics; classify data, data sources, and data types; explain the data analytics process and types of analytics; demonstrate fundamental MS Excel functions for data entry, formatting, formulas, and functions.	CLO1	I	I	I	
4-6	Explain data collection and preparation; assess data quality and apply ethical and legal considerations (RA 10173); import and clean data; apply descriptive analytics, including hypothesis tests and t-tests, to a legal or management dataset.	CLO2		E	E	I
7-11	Explain the importance of graphs and charts in data-driven decision-making; visualize caseload, compliance, and organizational trends; build interactive dashboards using PivotTables, slicers, and PivotCharts; communicate legal and management insights.	CLO3	E	E	D	E
13-17	Explain the principles of predictive analytics; perform regression and time-series forecasting; explain prescriptive analytics and apply what-if analysis and optimization to a legal or management problem; recommend data-driven actions ethically and responsibly.	CLO4	D	D	D	D

Legend: I - Introduced; E - Enabled; D - Demonstrated. The I-E-D levels show the progression from exposure, to guided practice, to evidence of performance.

ALIGNMENT MATRIX 2. CLO - GA - IO MAPPING

Course Learning Outcomes	Aligned Graduate Attributes	I-PEN Cluster	Institutional Outcomes	Level
CLO1	GA1, GA2	Innovator	IO1	I
CLO2	GA2, GA4	Innovator / People	IO1, IO2	E
CLO3	GA1, GA2	Innovator	IO1	D
CLO4	GA1, GA2, GA7	Innovator / Entrepreneur	IO1, IO3	D

TARGET SUSTAINABLE DEVELOPMENT GOALS (SDGS)

SDG 4 - Quality Education	Builds data literacy and digital skills that make legal management graduates future-ready professionals.
SDG 8 - Decent Work and Economic Growth	Supports data-driven decisions that promote sound organizational governance, productivity, and compliance.
SDG 9 - Industry, Innovation, and Infrastructure	Integrates analytics tools and technology to innovate legal and management processes.
SDG 16 - Peace, Justice, and Strong Institutions	Upholds ethics, the rule of law, data privacy (RA 10173), and accountability in handling legal and organizational data.
SDG 17 - Partnerships for the Goals	Supports evidence-based collaboration among legal, business, and government stakeholders through shared data and reporting.

COURSE OBJECTIVE

This course aims to develop students' ability to explain the fundamentals of data science and analytics; collect and prepare legal and management data with due regard for ethics, confidentiality, and data privacy; apply descriptive, predictive, and prescriptive analytics and data visualization using Microsoft Excel; and communicate data-driven recommendations that support sound legal and management decisions, consistent with the legal and management orientation of the BS Legal Management program.

COURSE ASSESSMENT AND ATTAINMENT MATRIX

CLO	Assessment Evidence	Measure	Performance Indicator	Rubric / Standard	Target	Type
CLO1	Quizzes, laboratory activity on Excel fundamentals, individual performance task, midterm exam items	Direct	Differentiates data science and analytics, explains the analytics process, and performs data entry, formulas, and functions accurately	75% mastery (discrete item scoring) or 3/4 Proficient (laboratory rubric)	>=75% of students meet threshold	Formative + Summative
CLO2	Laboratory activity on data cleaning and descriptive analytics, individual performance task, quiz, midterm exam items	Direct	Prepares and cleans a dataset (duplicates, missing values, outliers), applies descriptive statistics correctly, and observes data ethics, confidentiality, and privacy	3/4 Proficient in accuracy, completeness of cleaning, and ethical handling of data	>=75% of students meet threshold	Performance-based
CLO3	Chart and dashboard laboratory outputs, individual performance task, dashboard presentation, midterm exam items	Direct	Creates appropriate visualizations and an interactive dashboard that communicate legal and management insights	3/4 Proficient in chart selection, accuracy, interactivity, and clarity of insight	>=75% of students meet threshold	Performance-based
CLO4	Regression, forecasting, and what-if laboratory outputs; final analytics project; final examination	Direct	Builds and interprets predictive and prescriptive models and proposes justified, data-driven recommendations	3/4 Proficient in model accuracy, interpretation, and justification of recommendations	>=75% of students meet threshold	Summative

Note: Indirect validation measures may include course evaluation, self-assessment, and other relevant feedback. These help improve the course, but direct assessment evidence shall remain the primary basis for determining attainment.

COURSE PLAN

Weeks	Aligned CLO	Lesson Intended Learning Outcomes (LILOs)	Learning Content	Teaching-Learning Strategies	Resources	Assessment Tasks	Expected Output
1	Orientation	1. State the vision, mission, goals, and institutional outcomes of ISU. 2. Identify the content, requirements, and learning outcomes of the course and relate them to the program outcomes for BS Legal Management. 3. Practice the rules and regulations of the university, including the Classroom AI Policy.	University orientation; VMGO, quality policy, and institutional outcomes; overview of the course, CLOs, and program outcomes; syllabus discussion; course requirements and expectations; Classroom AI Policy	Interactive orientation, group discussion, group essay writing	Student manual, syllabus, institutional issuances; presentation of IO, VMGO, PO, and CLO	Recitation; group essay on the VMGO, quality policy, and institutional outcomes	Group essay; course understanding
2-3	CLO1	1. Differentiate data science and data analytics. 2. Classify data, data sources, and data types commonly used in legal and management settings (e.g., case records, compliance logs, contracts, client or organizational data). 3. Explain the data analytics process and the types of analytics. 4. Demonstrate proficiency in fundamental MS Excel functions: data entry, formatting, formulas, and functions.	Unit I. Overview of Data, Data Science Analytics, and Tools: about data science; about data analytics and its types; data, data sources, and data types; the data analytics process; applications in legal practice and management; MS Excel as a data analytics tool (interface, creating and saving workbooks, worksheets and data entry, formulas and functions)	Discussion and demonstration; hands-on laboratory exercises on data entry, formulas, and functions using sample legal and management data	Nandi & Sharma (2020); sample legal and management datasets; computer laboratory with MS Excel	Quiz; individual performance task; laboratory activity (hands-on exercise)	Formatted Excel workbook with formulas and functions
4-6	CLO2	1. Explain the importance of data collection and preparation. 2. Discuss basic data quality assessment and the ethical and legal considerations in data collection, including confidentiality and the Data Privacy Act of 2012 (RA 10173). 3. Import data from external sources into Excel and perform basic data cleaning, including removing duplicates and handling missing values and outliers. 4. Explain the key principles of descriptive analytics and apply them to a dataset using MS Excel.	Unit II. Data Collection and Preparation; Descriptive Analytics: importance of data collection and preparation; basic data quality assessment; confidentiality, ethical, and legal considerations in data collection and data privacy; importing data from various sources; data cleaning and preprocessing in Excel; handling missing data and outliers; overview of descriptive analytics; descriptive statistics (mean, median, standard deviation, etc.); Excel functions for data analysis; Data Analysis ToolPak; hypothesis tests and t-tests	Discussion and demonstration; case discussion on data privacy, confidentiality, and ethics; hands-on data cleaning and descriptive analysis of a legal or management dataset	Martino (2019); RA 10173 (Data Privacy Act of 2012); public legal and governance statistics (e.g., PSA, DOJ, SEC); computer laboratory	Quiz; individual performance task; laboratory activity (data cleaning and descriptive analytics)	Cleaned dataset and descriptive analytics summary (Excel workbook)
7-11	CLO3	1. Explain the importance of graphs and charts and their role in data-driven decision-making. 2. Create appropriate charts (bar, line, heatmap, histogram, box plot, scatter plot) to visualize distributions and trends in caseload, compliance, and organizational data. 3. Build interactive dashboards using PivotTables, slicers, filters, and PivotCharts that provide a comprehensive view of legal and management data. 4. Communicate insights from visualizations and dashboards orally and in writing.	Unit III. Data Exploration, Visualization, and Dashboards: the power of graphs and charts; data-driven decision-making; creating charts and graphs to visualize data distributions and trends; bar charts, line charts, heatmaps; histograms, box plots, scatter plots; advanced data visualization techniques in Excel (creating interactive dashboards, PivotTable slicers and filters, visualizing data with PivotCharts)	Discussion and demonstration; hands-on production of visualizations and interactive dashboards from legal and management datasets (e.g., case disposition rates, compliance violations, contract turnaround time, client intake trends); short dashboard presentations	Martino (2019); sample datasets; computer laboratory	Quiz; individual performance task; laboratory activity; dashboard presentation	Set of charts and an interactive Excel dashboard
12	CLO1-CLO3	Demonstrate mastery of Units I to III through a written and hands-on MS Excel examination.	Coverage: Weeks 2 to 11 (Units I to III)	Proctored written examination (lecture) and practical examination in the computer laboratory	Examination questionnaire, datasets, computer laboratory	Midterm Examination (ME)	Accomplished examination and Excel file
13-17	CLO4	1. Explain the key principles of predictive analytics and perform simple and multiple regression in Excel. 2. Analyze time-series data and develop forecasts (e.g., monthly case filings or compliance incidents) using Excel. 3. Explain the key principles of prescriptive analytics and apply What-If Analysis and optimization to a legal or management problem. 4. Conduct simulation and optimization based on a predictive model, and present data-driven recommendations ethically and responsibly.	Unit IV. Predictive and Prescriptive Analytics: overview of predictive analytics; simple linear regression; multiple regression; time series analysis (analyzing time-based data, creating time series charts and forecasting models); prescriptive analytics (overview, What-If Analysis tool, application of optimization to organizational problems); responsible and ethical use of predictive and prescriptive models	Discussion and demonstration; hands-on modeling, forecasting, and what-if exercises; case-based final analytics project	Fávero & Belfiore (2019); Klimberg (2023); Roque (2022); Tutorialspoint (2016)	Quiz; individual performance task; laboratory activity; final analytics project	Regression and forecasting workbook; final analytics project (dashboard, forecast, and recommendation)
18	CLO4	Demonstrate mastery of the course learning outcomes, with emphasis on Unit IV, through a written and hands-on MS Excel examination.	Coverage: Unit IV, with integration of Units I to III	Proctored written examination (lecture) and practical examination in the computer laboratory	Examination questionnaire, datasets, computer laboratory	Final Examination (FE)	Accomplished examination and Excel file

Time allocation: 18 weeks at 1 lecture hour and 3 laboratory hours per week (18 lecture hours and 54 laboratory hours in total).

COURSE CQI/PDCA MECHANISM

PLAN	DO	CHECK / ASSESS	ACT
Align CLOs with approved Program Outcomes, Graduate Attributes, and Institutional Outcomes; identify assessment evidence and attainment targets.	Implement the course plan, administer learning activities and assessments, and collect direct evidence of student performance.	Compute CLO attainment using agreed metrics such as percentage of students meeting the mastery threshold and mean rubric score per CLO.	Use the attainment summary, gap analysis, and student feedback to revise content, strategies, assessment tools, and support mechanisms in the next course offering.

Minimum attainment standard: A Course Learning Outcome is considered attained if at least 75% of students meet the mastery threshold, such as 75% score or Proficient / Level 3 in rubric-based tasks.

CLASSROOM AI POLICY FOR STUDENT ASSESSMENT (BOR NO. 66, SERIES OF 2026)

The acceptable AI assistance threshold for student assessment submissions, including but not limited to case studies, narrative reports, capstone or special projects, theses, dissertations, reaction papers, and reflection papers, shall not exceed 25%.

Minor Violations (26% to 50% similarity)	Faculty are expected to guide students through constructive discussions about responsible AI use, issue verbal warnings, and provide opportunities to redo assessment tasks. The faculty also takes a proactive educational approach, ensures students understand institutional expectations, and requires a promissory letter if the student receives three (3) verbal warnings.
Moderate Violations (51% to 75% similarity)	Faculty move toward more formal interventions, such as organizing official meetings and imposing grade reductions on the assessment task.
Severe Violations (76% to 100% similarity)	Faculty respond with decisive action by assigning a failing grade for the assessment task, coordinating student counseling with college services, and emphasizing the seriousness of the violation.

GRADING CRITERIA

Course with Laboratory

Lecture (60%)		
Formative Assessment		50%
Summative Assessment		
• Midterm Examination (ME)		25%
• Final Examination (FE)		25%
	Total (Lec)	**100%**
Laboratory (40%)		
Formative Assessment		50%
Summative Assessment		
• Midterm Examination (ME)		25%
• Final Examination (FE)		25%
	Total (Lab)	**100%**

Grade= [TOTAL (Lec) *0.6 + [TOTAL (Lab)]*0.4

GRADE EQUIVALENT

98.00 – 100.00	**1.00**
95.00 – 97.00	**1.25**
92.00 – 94.00	**1.50**
89.00 – 91.00	**1.75**
86.00 – 88.0	**2.00**
83.00 – 85.00	**2.25**
80.00 – 82.00	**2.50**
77.00 – 79.00	**2.75**
75.00 – 76.00	**3.00**
Below 75.00	**5.00**

CLASSROOM POLICIES

1. Students are expected to follow University rules on attendance, dress code, decorum, and professional conduct.
2. Cheating, plagiarism, academic dishonesty, and other forms of misconduct are strictly prohibited.
3. Use of gadgets and digital platforms is allowed only for learning purposes and when permitted by the instructor.
4. Safety, orderliness, courtesy, and respect for diversity shall be observed at all times.
5. For complete guidelines, students shall refer to the latest approved Student Manual and related University issuances.

REFERENCES

- Fávero, L. P., & Belfiore, P. (2019). Data science for business and decision making. Academic Press.
- Klimberg, R. (2023). Fundamentals of predictive analytics with JMP. SAS Institute.
- Martino, J. C. R. (2019). Hands-On Machine Learning with Microsoft Excel 2019: Build complete data analysis flows, from data collection to visualization. Packt Publishing Ltd.
- Nandi, G., & Sharma, R. K. (2020). Data Science fundamentals and practical approaches: understand why data science is the next. BPB Publications.
- Roque, F. (2022). Data Forecasting and Segmentation Using Microsoft Excel: Perform Data Grouping, Linear Predictions, and Time Series Machine Learning Statistics Without Using Code. Packt Publishing Limited.
- Tutorialspoint. (2016). Data Analysis with MS Excel. Retrieved from https://www.tutorialspoint.com/excel_data_analysis
- Commission on Higher Education. (2012). CHED Memorandum Order No. 46, Series of 2012: Policy-standard to enhance quality assurance in Philippine higher education through an outcomes-based and typology-based quality assurance system.
- Republic Act No. 10173 (Data Privacy Act of 2012).
- Isabela State University Outcomes-Based Education Framework.
- Latest approved ISU Student Manual, OBE guidelines, and relevant issuances.

DOCUMENT REVISION HISTORY

Revision No.	Date	Description of Revision	Basis/Remarks	Prepared/Updated by	Reviewed by	Approved by
1.0	July 13, 2026	Initial enhanced OBE Course Syllabus Template for presentation and review	For University-wide review and adoption	Office of the VPAA/Director for Instruction	Academic Affairs Directors, Deans, and Program Chairs	Vice President for Academic Affairs
2.0	July 14, 2026	Major revision and enhancement of the OBE Course Syllabus Template	Based on the consultation with the Academic Affairs Office and the Deans and alignment with the University OBE Framework, CQI, and accreditation requirements	Office of the VPAA/Director for Instruction	Academic Affairs Directors, Deans, and Program Chairs	Vice President for Academic Affairs

Prepared by	Reviewed by	Approved by	Date
APRIL JOY G. LUNG-AYAN, LPT Instructor	_____ Program Chair	_____ Dean, College of _____	_____